

Algebraic Geometry in Chromatic Homotopy Theory

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Abstract

This is a note of Chromatic homotopy theory, from the perspective of Algebraic Geometry.

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1 Stable Homotopy

1.1 Localization

We denote $\mathcal{Spc}$ as the homotopy category of spectra, and let $E \in \mathcal{Spc}$,

Definition 1.1. A spectrum X is E -acyclic if $E \wedge X \simeq 0$. A morphism $f : X \rightarrow Y$ in $\mathcal{Spc}$ is a E -equivalence if $E \wedge f : E \wedge X \rightarrow E \wedge Y$ is an equivalence.

A spectrum X is E -local if the following two equivalent condition are satisfied:

1. for E -equivalence $f : Y \rightarrow Z$, $[f, X] : [Z, X] \rightarrow [Y, X]$ is an isomorphism
2. for E -acyclic F , $[F, X] \cong 0$

Lemma 1.1. The two conditions are indeed equivalent.

Proof. $1 \implies 2$ We have $0 \rightarrow F$ is a E -equivalence by assumption, so $[F, X] \rightarrow [0, X] \cong 0$ is an isomorphism.

$2 \implies 1$ If $f : Y \rightarrow Z$ is a E -equivalence, then cofiber cof of f will be E -acyclic, as $\wedge$ preserve colimit, hence we have f induces isomorphism $[Z, X] \rightarrow [Y, X]$ $\square$

The essential full subcategory of $\mathcal{Spc}$ whose objects are the E -local spectra will be denoted as $\mathcal{Spc}_E$.

It is not hard to see that $\mathcal{Spc}_E$ is closed under homotopy limits of $\mathcal{Spc}$.

Proposition 1.2. If $f : X \rightarrow Y$ is a E -equivalence between E -local spectra, then f is an equivalence.

Proof. Take the fiber of f , F , which is E -acyclic, and it will also be E -local. Suppose Z is a E -local and -acyclic spectrum then $[Z, Z] \cong 0$, so $id_Z \simeq 0$. In other word, $Z \simeq 0$. Hence f is an equivalence. $\square$

Definition 1.2. For any spectrum X , a E -equivalence $\varphi : X \rightarrow L_EX$, where L_EX is E -local, is called a E -localization.

Proposition 1.3. If exists, φ is unique up to equivalence.

Proof. This comes from the fact E -equivalence is an equivalence between E -local spectra. $\square$

Suppose E is an multiplicative ring spectrum (the weakest assumption that we can have), for any E -module M , we have $(M \xrightarrow{e_M \wedge id} E \wedge M \xrightarrow{\varphi} M) \simeq id$ For any $f : Y \rightarrow M$, there are commutative diagram

$$\begin{array}{ccc} Y & \xrightarrow{f} & M \\ \downarrow e_M \wedge id & & \downarrow e_M \wedge id \\ E \wedge Y & \xrightarrow{E \wedge f} & E \wedge M \end{array}$$

So if Y is E -acyclic, then f factors through 0. Therefore, M is E -local.

Definition 1.3. For any $X \in \text{Spc}$, let $E_\bullet X := E^{\wedge \bullet + 1} \wedge X$ be the cosimplicial spectrum that comes from the multiplicative structure on E , then $E_n X$ are all E -local, thus $\text{Tot}(E_\bullet X)$ is also E -local. There is $X \rightarrow \text{Tot}(E_\bullet X)$.

For many cases of E , this will be a E -equivalence, hence a E -localization of X .

When we see totalization, we think spectral sequence, and indeed we have spectral sequence. For E a flat commutative ring spectrum, $(E_*, E_* E)$ admits a Hopf algebra structure, and we have the Adams-Novikov Spectral Sequence

$$E_2^{s,t} = \text{Ext}_{E_* E}^{s,t}(E_*, E_*(X)) \implies \pi_{t-s} L_E X$$

That converges for spectra such that $X = S$.

1.2 Complex Oriented Cohomology

Definition 1.4. A complex orientable spectrum is a multiplicative spectrum E such that $\tilde{E}^2(\mathbb{C}P^\infty) \rightarrow \tilde{E}^2(\mathbb{C}P^1) \cong E^0$ is surjective.

Since E is multiplicative E^0 has generator, which correspond to a generator $t \in E^2(\mathbb{C}P^1)$. There is then an element $t \in E^2(\mathbb{C}P^\infty)$.

We have a map $E^*[t]/(t^{n+1}) \rightarrow E^*(\mathbb{C}P^n)$, from the fact that there is a cover of $n+1$ contractible open set of $\mathbb{C}P^n$. As it turns out

Lemma 1.4. [2, Lemma 1.4] The map is an isomorphism.

Proof. For $E = H\mathbb{Z}$, this is a classic result. We generalize it using the Atiyah-Hirzebruch SS, which is multiplicative, with one generator t that correspond to the element $t \in E^2(\mathbb{C}P^n)$. So the SS collapse on the E_2 page and all is done. $\square$

We can take one step further, from Milnor's sequence

$$0 \rightarrow \lim^1 E^{*-1}(\mathbb{C}P^n) \rightarrow E^*(\mathbb{C}P^\infty) \rightarrow \lim E^*(\mathbb{C}P^n) \rightarrow 0$$

and the fact $E^*(\mathbb{C}P^{n+1}) \rightarrow E^*(\mathbb{C}P^n)$ is surjective, we have

$$E^*(\mathbb{C}P^\infty) \cong E^*[[t]]$$

The name complex orientation has a geometric meaning. We know complex line bundle is classified by the space $\mathbb{C}P^\infty \simeq BU(1)$. Given a complex oriented cohomology, t is then a characteristic class of universal complex line bundle.

The tensor product of line bundle $\mathbb{C}P^\infty \times \mathbb{C}P^\infty \rightarrow \mathbb{C}P^\infty$ then induces a map

$$\psi_E : E^*(\mathbb{C}P^\infty) \cong E^*[[t]] \rightarrow E^*(\mathbb{C}P^\infty \times \mathbb{C}P^\infty) \cong E^*[[x, y]]$$

The generator t then correspond to a formal power series in x, y , $F_E(x, y) := \psi_E(t)$.

Definition 1.5. A formal group law over a ring R is a formal power series $F(x, y)$ such that

- $F(x, 0) = F(0, x) = x$
- $F(x, y) = F(y, x)$
- $F(x, F(y, z)) = F(F(x, y), z)$

For R graded, we require $F(x, y)$ to be homogeneous of degree 2.

We will write $x +_F y := F(x, y)$.

Lemma 1.5. [8, Proposition 6.2 Chapter VII] Let E be a complex oriented cohomology, $F_E(x, y)$ is a graded formal group law.

Proof. This follows from the properties of tensor product of line bundle being unital, associative and commutative. $\square$

Example 1.6. Let $E = HR$ for some ring R , $f_E(x, y) = x + y$, as it is the only possible graded formal group law.

Let $E = K$ be the complex K -theory, we choose $t \in K^2(\mathbb{C}P^\infty)$ to be the one that correspond to $[\mathcal{O}(1)] - 1$, where $\mathcal{O}(1)$ is the universal line bundle and 1 is the trivial rank 1 bundle over $\mathbb{C}P^\infty$, under the bott map $\beta : K^* \rightarrow K^{*+2}$. We have that given line bundle

$$[\mathcal{O}(1)] \otimes [\mathcal{O}(1)] - 1 = ([\mathcal{O}(1)] - 1) \otimes ([\mathcal{O}(1)] - 1) + ([\mathcal{O}(1)] - 1) \otimes 1 + 1 \otimes ([\mathcal{O}(1)] - 1)$$

So $F_K(x, y) = x + y + \beta^{-1}xy$

1.3 Universal Complex Oriented Cohomology

The inclusion

$$BU(1) \hookrightarrow BU(2) \hookrightarrow \dots \hookrightarrow BU(n)$$

Assembles into the inclusion of Thom spaces of the universal rank n bundle, that is $\Sigma^2 MU(n-1) \rightarrow MU(n)$. Define

$$MU := \lim_n \Sigma_+^{\infty-2n} MU(n)$$

This spectrum has many interesting property,

Proposition 1.7. MU is complex oriented, moreover commutative.

Proof. We let the complex orientation η to be $\Sigma_+^{\infty-2} MU(1) = \Sigma_+^{\infty-2} BU(1) \rightarrow MU$ to be our complex orientation. The multiplication is induced from the direct sum of complex bundle $BU(n) \times BU(m) \rightarrow BU(n+m)$, and commutativity comes from that direct sum is commutative. $\square$

Let E be multiplicative, we see that given a ring map $\psi : MU \rightarrow E$ determines a complex orientation $\psi(\eta)$, and from a commutative complex oriented cohomology (E, t) , we can construct a ring map $MU \rightarrow E$ unique up to homotopy. Therefore, we have

Theorem 1.8. *[4, Lecture 6] There is a bijection between homotopy class of ring maps $MU \rightarrow E$ where E is commutative, and complex orientation on E .*

So we have that commutative complex oriented cohomology is just a MU -ring spectrum. This justifies the title of this section, universal complex oriented cohomology.

Where does graded formal group law fit in this discussion? Well this comes from the classic result of Lazard.

As a soft invitation to algebraic geometry, we denote $\mathcal{A}ff$ the category of Affine schemes, equivalently the opposite category of commutative ring.

Definition 1.6. *The functor $\mathbf{fgl} : \mathcal{A}ff^{op} \rightarrow \mathbf{Sets}$ takes commutative ring R to the set of formal group law over R , and given morphism $\psi : R \rightarrow R'$, $\mathbf{fgl}(\psi)$ takes formal group law $F(x, y) = \sum c_{ij}x^i y^j$ to $\mathbf{fgl}(\psi)(F(x, y)) = \sum \psi(c_{ij})x^i y^j$.*

It is easy to see that, since ring morphism $\mathbb{Z}[c_{ij}] \rightarrow R$ determines a power series $F(x, y)$, so the functor $\mathbf{fgl}$ is represented by the ring $L' = \mathbb{Z}[c_{ij}]/(R)$, where R is some relation on c_{ij} a formal group law must satisfy, and it has the universal formal group law

$$F_U(x, y) = \sum c_{ij}x^i y^j$$

We define polynomial

$$C_n(x, y) := \frac{1}{d}[(x + y)^n - x^n - y^n]$$

Here $d = p$ if $n = p^k$ and $d = 1$ else. It is not hard to see that C_n is homogeneous of degree $2n$.

Let $L = \mathbb{Z}[a_1, a_2, \dots, a_n, \dots]$, there exists a formal group law F such that

$$F(x, y) \equiv x + y + \sum_{n=1}^{\infty} a_n C_{n+1}(x, y) \pmod{(a_1, \dots, a_n, \dots)^2}$$

The morphism $L' \rightarrow L$ classifying F determines an isomorphism and we have that $\mathbf{fgl}$ is represented by $\text{spec}(L)$.

Since given a complex orientation $MU \rightarrow E$, we have a graded ring map $MU^* \rightarrow E^*$ that determines a graded formal group law. Therefore, there exists a map

$$L^* := \mathbb{Z}[a_1, a_2, \dots, a_n, \dots] \rightarrow MU^*, |a_n| = -2n$$

Theorem 1.9. *(Quillen) The map above is an isomorphism.*

Here we use the grading where $\pi_* MU$ is positively graded, hence MU^* is negatively graded.

2 Formal Groups

2.1 Algebraic Geometry

We will talk about the type of morphisms that we will see, there are many other as well that are important.

Definition 2.1. An open immersion of scheme is a morphism $f : X \rightarrow Y$ such that it is a homeomorphism onto an open set of Y , and $f^{-1}\mathcal{O}_Y \rightarrow \mathcal{O}_X$ is an isomorphism.

An affine morphism of scheme is a morphism $f : X \rightarrow Y$ such that for every $\text{spec}(R) \rightarrow Y$, $\text{spec}(R) \times_Y X$ is affine.

A closed embedding of scheme is a morphism $f : X \rightarrow Y$ such that it is a homeomorphism onto a closed subset of Y and $\mathcal{O}_Y \rightarrow f_*\mathcal{O}_X$ is surjective.

A morphism $f : X \rightarrow Y$ is separated if the diagonal $\Delta : X \rightarrow X \times_Y X$ is a closed embedding.

Remark 2.1. The name open immersion is geometrically clear, as a sheaf of functions over an open subset of a geometric space should be just the restriction.

The name closed embedding is similar. However, the closedness tells us that the restriction of functions over Y to X would lose information on the edge.

Separated morphism is a relative Hausdorffness of schemes. The topology on schemes are never (almost) Hausdorff, think $\mathbb{A}_{\mathbb{C}}^1$. Therefore, we translate the language of topology into complete geometry.

If we translate the language into differential geometry, an affine morphism should be a smooth map such that preimage of locally euclidean subset into euclidean subsets. Such as vector bundle, open immersion, closed embedding (yes, it is not clear but they are affine).

Definition 2.2. A $\mathcal{O}_X$ -module $\mathcal{M}$ is quasicoherent if there is an open cover U of X and exact sequences

$$\bigoplus_j \mathcal{O}_X|_U \rightarrow \bigoplus_j \mathcal{O}_X|_U \rightarrow \mathcal{M}|_U$$

Let $\mathcal{Q}\text{coh}_X$ be the category of quasicoherent modules over X . For $f : X \rightarrow Y$, there exists adjoint pair

$$f^* : \mathcal{Q}\text{coh}_Y \rightarrow \mathcal{Q}\text{coh}_X : f_*$$

Here f_* takes $\mathcal{O}_X$ module $\mathcal{M}$ to the abelian sheaf $f_*\mathcal{M}$ over Y with $\mathcal{O}_Y$ -structure from $\mathcal{O}_Y \rightarrow f_*\mathcal{O}_X$. We refer to it as the pushforward. Here f^* takes $\mathcal{O}_Y$ module $\mathcal{N}$ to $\mathcal{O}_X \otimes_{f^{-1}\mathcal{O}_Y} f^{-1}\mathcal{N}$. We refer to it as the pullback.

For a closed embedding $f : X \rightarrow Y$, the kernel of $\mathcal{O}_Y \rightarrow f_*\mathcal{O}_X$ is then a quasicoherent module, the ideal $\mathcal{I} \subseteq \mathcal{O}_Y$.

Remark 2.2. Recall that I said closed embedding are affine, this comes from the preimage affine cover $\text{spec}(R) \subseteq Y$ looks like $\text{spec}(B/I)$, where I is the ideal $\mathcal{I}(\text{spec}(R))$

We fix a base scheme $S \in \text{Sch}$

Definition 2.3. Let $f : X \rightarrow S$, a $f^{-1}\mathcal{O}_S$ -derivation is a $f^{-1}\mathcal{O}_S$ -linear map $D : \mathcal{O}_X \rightarrow \mathcal{M}$ such that for any $a, b \in \mathcal{O}_X(U)$ for any $U \subseteq X$.

$$D(ab) = aD(b) + bD(a)$$

The Kahler differential $\Omega_{X/S}$ for $X \rightarrow S$ is the quasicoherent $\mathcal{O}_X$ -module that classifies $f^{-1}\mathcal{O}_S$ -derivations, that is there is a module map $d : \mathcal{O}_X \rightarrow \Omega_{X/S}$, such that

$$\text{Hom}_{\mathcal{O}_X}(\Omega_{X/S}, \mathcal{M}) \rightarrow \text{Der}_{f^{-1}\mathcal{O}_S}(\mathcal{O}_X, \mathcal{M})$$

is an isomorphism.

There is a nice formula to compute $\Omega_{X/S}$ when $X \rightarrow S$ is separated. Let $\mathcal{J} \subseteq \mathcal{O}_{X \times_S X}$ be the ideal defining the embedding $\Delta : X \rightarrow X \times_S X$, then

$$\Omega_{X/S} \cong \Delta^*(\mathcal{J}/\mathcal{J}^2)$$

For S -scheme morphism $i : X \rightarrow Y$, there is an exact sequence

$$i^*\Omega_{Y/S} \rightarrow \Omega_{X/S} \rightarrow \Omega_{X/Y} \rightarrow 0$$

For any closed embedding $i : X \rightarrow Y$, let $\mathcal{J}_X \subseteq \mathcal{O}_Y$ be the ideal defining i , there is an exact sequence

$$i^*(\mathcal{J}_X/\mathcal{J}_X^2) \rightarrow i^*\Omega_{Y/S} \rightarrow \Omega_{X/S} \rightarrow 0$$

Definition 2.4. [1, Definition 1.25] The module

$$\omega_i := i^*(\mathcal{J}_X/\mathcal{J}_X^2)$$

is called the conormal sheaf of the embedding $i : X \rightarrow Y$.

After we finished the local talk on geometry, let's talk globally.

Definition 2.5. Fix a base scheme $S \in \text{Sch}$, the Zariski topology Zar on $\text{Aff}/_S$ is generated by cover $\{X_i \rightarrow X\}$ such that $X_i \rightarrow X$ are open immersion and

$$\coprod_i X_i \rightarrow X$$

is surjective.

A scheme represents a presheaf $\text{Hom}_{\text{Sch}}(-, X)$, which is Zariski.

Definition 2.6. A morphism of scheme $f : X \rightarrow Y$ is flat if for every $x \in X$

$$\varphi_f : \mathcal{O}_{Y, f(x)} \rightarrow \mathcal{O}_{X, x}$$

is flat. A flat morphism of scheme $f : X \rightarrow Y$ is faithfully flat if the underlying continuous map is surjective.

A space X is quasicompact (or just compact depend how one learn it) if all open cover admits a finite subcover. A continuous map $f : X \rightarrow Y$ is quasicompact if for every quasicompact open set $V \subseteq Y$, $V \times_X Y$ is quasicompact.

A morphism of scheme is quasicompact if the underlying continuous map is quasicompact.

Definition 2.7. Fix a base scheme $S \in \text{Sch}$, the fpqc topology of $\text{Aff}/_S$ is generated by cover $\{X_i \rightarrow X\}$ such that $\coprod_i X_i \rightarrow X$ is faithfully flat and quasicompact.

We have that a Zariski cover is obviously fpqc, so fpqc is a subcanonical topology.

The fpqc topology is important to homotopy theorist for its appearance in both algebraic geometry and stable homotopy. More on this later.

We may define everything in terms of sites.

Given a scheme X , we define the structure sheaf $\mathcal{O}_X : \text{Aff}/_X^{op} \rightarrow \text{Ring}$ to be the sheaf of ring respect to the fpqc-topology such that on the affines

$$\mathcal{O}_X(\text{spec}(R) \rightarrow X) = R$$

Definition 2.8. A cartesian module over X is a fpqc-sheaf of $\mathcal{O}_X$ -module $\mathcal{M} : \text{Aff}/_X^{op} \rightarrow \text{Ab}$ such that for affine $\text{spec}(B) \rightarrow \text{spec}(A)$ over X

$$B \otimes_A \mathcal{M}(\text{spec}(A) \rightarrow X) \rightarrow \mathcal{M}(\text{spec}(B) \rightarrow X)$$

are an isomorphisms of B -module.

Let Qcoh_X be the category of quasicoherent module over X and let Mod_X^{cart} be the category of cartesian module over X . There exists adjoint pair

$$\epsilon^* : \text{Qcoh}_X \rightarrow \text{Mod}_X^{cart} : \epsilon_*$$

Here ϵ^* is defined by pullback, so for $\epsilon^*\mathcal{M}(f : Y \rightarrow X) = f^*\mathcal{M}(Y)$, and $\epsilon_*\mathcal{N}(U) = \mathcal{N}(U \hookrightarrow X)$.

Note this is indeed defined. If $\mathcal{M}$ is qcqh, then the pullback along $f : \text{spec}(A) \rightarrow X$ is a qcqh $\mathcal{O}_{\text{spec}(A)}$ -module. Since qcqh module over affine is equivalent to the module of the structure ring under taking the section functor, we have that ϵ^* is cartesian. Vice versa, if $\mathcal{N}$ is cartesian, take some affine cover of X , we can easily see that $\epsilon_*\mathcal{N}$ is qcqh. Moreover, this is actually an equivalence of category. Therefore, we may safely transfer between cartesian modules and quasicoherent modules.

2.2 Formal Lie Variety and Formal Group

Definition 2.9. Let $i : X \rightarrow Y$ be a closed embedding of schemes and let $\mathcal{I}_X$ be the defining ideal, we define the n^{th} infinitesimal neighbourhood $\text{Inf}_X^n(Y) \subseteq Y$ to be the closed subscheme of Y defined by $\mathcal{I}_X^{n+1}$. There is $\text{Inf}_X^n(Y) \rightarrow \text{Inf}_X^{n+1}(Y)$

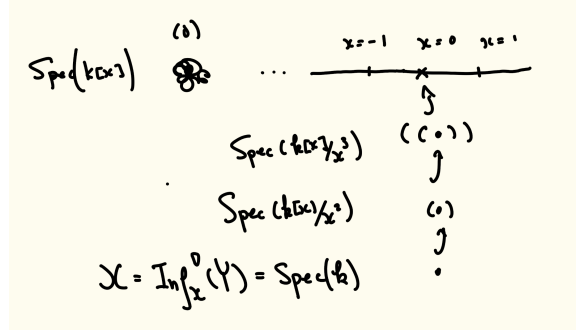


Figure 1:

A good picture for this would be the inclusion of the point $x = 0$ $\text{spec}(k) \rightarrow \text{spec}(k[x])$. The ideal of definition is obviously (x) , and therefore, $\text{Inf}_X^n(Y) = \text{spec}(k[x]/(x^{n+1}))$. I like to picture them as Figure 1

The definition generalize to all *fpqc*-sheaves over a fixed base scheme S . Given injection of *fpqc*-sheaves $i : X \rightarrow Y$, let $\text{Inf}_X^n(Y) \subseteq Y$ to be the subsheaf such that on quasicompact $U \rightarrow S$, $[\text{Inf}_X^n(Y)](U)$ is the set of sections a in $Y(U)$, such that there is a *fpqc*-cover $V \rightarrow U$ and a closed subscheme of $V' \subseteq V$ with ideal of definition vanishing at $n + 1$ power, and

$$a|_{V'} \in Y(V')$$

belongs to $X(V')$. (Let's ignore how it generalizes and just take it for granted)

Definition 2.10. [10, Definition 4.1] A formal S -scheme is a filtered colimit of schemes $X_\bullet : I \rightarrow \text{Sch}/_S$ in Zariski sheaves.

Example 2.3. Let R be an I -adic ring for some ideal $I \subseteq R$, that is $R \cong \lim_n R/I^n$. For instance $R = k[[x_1, \dots, x_n]]$ is $(x_1, \dots, x_n)$ -adic, or $\mathbb{Z}_p$ is $p\mathbb{Z}_p$ -adic.

We can equip R with the I -adic topology, the weakest topology induced by letting R/I^n be discrete.

We denote the formal scheme correspond to $\text{spec}(R/I^n)$ as $\text{spf}(R)$.

Definition 2.11. Let X *fpqc*-sheaf over S , and let $e : S \rightarrow X$ to be a section. X is ind-infinitesimal if $\text{colim}_n \text{Inf}_S^n(X) \rightarrow X$ is an isomorphism.

We assume that each of $X_n := \text{Inf}_S^n(X)$ to be representable, i.e. it is a S -scheme. This means that X is a formal S -scheme.

In this case, we define ω_e to be the conormal sheaf of any of the closed embedding $e_n : S \rightarrow X_n$, (notice the n is irrelevant). Of course, the exact sequence implies $\omega_e \cong e_n^* \Omega_{X_n/S}$.

Denote $\mathcal{I}_n$ to be the ideal of definition for the embedding $e_n : S \rightarrow X_n$. There is a graded $\mathcal{O}_S$ -algebra associated to the filtration on X . We define $\text{gr}_k(X)$ to be $e_n^*(\mathcal{I}_n^k/\mathcal{I}_n^{k+1})$ for some $n \geq k$

(once again, n doesn't matter). The algebra structure comes from

$$\begin{aligned} \mathrm{gr}_k(X) \otimes_{\mathcal{O}_S} \mathrm{gr}_{k'}(X) &= e_n^*(\mathcal{I}_n^k / \mathcal{I}_n^{k+1}) \otimes_{\mathcal{O}_S} e_n^*(\mathcal{I}_n^{k'} / \mathcal{I}_n^{k'+1}) \\ &= e_n^*(\mathcal{I}_n^k / \mathcal{I}_n^{k+1} \otimes_{\mathcal{O}_{X_n}} \mathcal{I}_n^{k'} / \mathcal{I}_n^{k'+1}) \\ &= e_n^*(\mathcal{I}_n^k / \mathcal{I}_n^{k+1} \otimes_{\mathcal{O}_{X_n}} \mathcal{I}_n^{k'} / \mathcal{I}_n^{k'+1}) \\ &\rightarrow e_n^*(\mathcal{I}_n^{k+k'} / \mathcal{I}_n^{k+k'+1}) \end{aligned}$$

For large enough n .

We see there is a natural map from $\omega_e^{\otimes k} \rightarrow \mathrm{gr}_k(X)$, which is symmetric.

Definition 2.12. *A formal Lie variety over S is an ind-infinitesimal fpqc-sheaf X with a section $e : S \rightarrow X$ such that*

- X_n are representable and affine over S
- ω_e is locally free of finite rank over S
- The morphism of graded algebra $\mathrm{Sym}_*(\omega_e) \rightarrow \mathrm{gr}_*(X)$ is an isomorphism.

A morphism of formal Lie variety $(X, e) \rightarrow (X', e')$ is a morphism of sheaves that takes e to e' .

Let's unravel the definition with geometric words. The section $e : S \rightarrow X$ is a point of S -scheme X . The first condition says that the local neighbourhood of e is affine (flat). ω_e is locally free of finite rank, which says that X is smooth around e , and it is finite dimensional around X . The last condition is a bit weird. We should think of it as comparing n^{th} -infinitesimal information determined by $\mathrm{gr}_*(X)$, with the polynomial of the 1^{st} infinitesimal information, which is controlled by ω_e . In a good space, they should equal.

Given a morphism of scheme $f : T \rightarrow S$, and a formal Lie variety (X, e) over S , (f^*X, f^*e) is a formal Lie variety over T .

Remark 2.4. *If $S = \mathrm{spec}(A)$, it can be shown that (X, e) is a formal Lie variety iff $X \cong \mathrm{spf}(A[[x_1, \dots, x_n]])$ [1, Remark 1.3.1]. Therefore, locally speaking, i.e. choose an affine cover of the general scheme S , there is a formal affine space describing the formal Lie variety.*

If X and Y are formal Lie variety over S , then $X \times_S Y$ is again another formal Lie variety [1, Lemma 2.1].

Definition 2.13. [1, Definition 2.2] *A formal group over S is a formal Lie variety (G, e) that is also a group object of formal Lie variety such that $\omega_G := \omega_e$ is rank 1.*

Remark 2.5. *For $S = \mathrm{spec}(A)$ we have that a formal group is then equivalent to a formal group law over A . A homomorphism of formal group $\phi : F \rightarrow F'$ is equivalent to a formal power series ϕ over A such that $\phi(x) +_{F'} \phi(y) = \phi(x +_F y)$. This would implies $\phi \in xA[[x]]$.*

For an arbitrary scheme S , find a cover $\mathrm{spec}(A_i)$ of S , a formal group over S is a set of formal group laws over A_i that are compatible along the intersection.

Remark 2.6. We know MU_* is equivalent to the Lazard ring, classifies the formal group laws, as it turns out, the MU_* -Hopf algebroid MU_*MU classifies the strict isomorphisms between formal group law. This is the isomorphisms of formal group law ϕ , whose derivative at 0, $\phi'(0) = 1$. So the coefficient in front of x of ϕ is 1.

We often like to talk about general isomorphisms of formal group, hence we would attach an invertible element u of grade 0 to MU_*MU , so the MU_* -Hopf algebroid $MU_*MU[u^{\pm}]$ would classify all isomorphisms.

When we talk about groupoid in schemes and how they determine algebraic stacks, we will have that the groupoid $(\text{spec}(MU_*), \text{spec}(MU_*MU[u^{\pm}]))$ determines the moduli stack of formal groups $\mathcal{M}_{\text{fg}}$.

2.3 Landweber Exact is Flatness over a Better Geometric Space

In order to keep us motivated with the geometric construction that we might or might not care, I'll start a discussion.

Suppose R is a MU_* -algebra, therefore has a formal group law over R (of grade -2 , because we flipped the grading), define $R_* : \mathcal{S} \rightarrow \mathcal{Ab}$ as

$$R_*(X) = MU_*(X) \otimes_{MU_*} R$$

Is this a homology?

Well, no, or our study of homology would be a little too boring. The key is that, even though we have for homotopy cofiber sequence $X \rightarrow Y \rightarrow Z \rightarrow \Sigma X$

$$\dots MU_*(X) \rightarrow MU_*(Y) \rightarrow MU_*(Z) \rightarrow MU_{*-1}(X) \rightarrow \dots$$

is exact, it might not be true for $R_*(-)$.

When is it true? Well for R is a flat MU_* -algebra, but this is actually too strong of a condition.

We write $[n]_F(x)$ to be the power series of $x +_F x +_F \dots +_F x$ n times. We have that for any prime p

$$[p]_F(x) = \sum_i a_i x^i$$

We denote $v_i = a_{p^i}$.

A sequence $r_0, r_1, \dots, r_n, \dots \in A$ is called regular for A -module M if r_0 is not a zero divisor of M , and r_n is not a zero divisor of $M/(r_0, \dots, r_{n-1})M$.

Theorem 2.7. (Landweber Exact Functor Theorem) $R_*(-)$ is a homology if for every prime p , the sequence $v_0 := p, v_1, \dots, v_n, \dots \in MU_*$ of the formal group law associated with R is regular in R .

We call R that satisfies the above condition to be Landweber exact. The statement generalizes for MU_* -modules.

As it turns out, Landweber exactness is just flatness, but for a better geometric object, the moduli stack of formal group.

Proposition 2.8. [5, Proposition 7] *A MU_* -algebra R_* is Landweber exact iff $\text{spec}(R_*) \rightarrow \text{spec}(MU_*) \rightarrow \mathcal{M}_{\text{fg}}$ is flat.*

Similarly for MU_* -module M , there is an associated quasicoherent module, $\mathcal{F}_M$ over $\mathcal{M}_{\text{fg}}$ (whatever quasicoherent modules are for stack, more on this later). Landweber exact M is equivalent to a flat $\mathcal{F}_M$ over $\mathcal{M}_{\text{fg}}$.

I think this is a good motivation for us to shift are focus from MU_* -module and simple algebra to actual hardcore algebraic geometry.

3 Stacks

3.1 Abstract Nonsenses

Definition 3.1. [9, 1.4.35] *Fix a category C , a category fibered in groupoids over C is a functor $p : P \rightarrow C$ such that*

- *For every $f : V \rightarrow U$ in C and a $x \in P$ such that $p(x) = U$, there exists $\tilde{f} : y \rightarrow x$ such that $p(\tilde{f}) = f$*

$$\begin{array}{ccc} \tilde{f} : y & \cdots \rightarrow & x \in P \\ \Downarrow p & & \Downarrow p \\ f : V & \longrightarrow & U \in C \end{array}$$

- *For any $\tilde{f} : y \rightarrow x$ and $\tilde{g} : z \rightarrow x$, if there is $h : p(y) \rightarrow p(z)$ such that $p(\tilde{g}) \circ h = p(\tilde{f})$, then h has an unique lift $\tilde{h} : y \rightarrow z$ in P such that $\tilde{g} \circ \tilde{h} = \tilde{f}$*

$$\begin{array}{ccc} & z & \\ \tilde{h} \nearrow & & \searrow \tilde{g} \\ y & \xrightarrow{\tilde{f}} & x \\ \Downarrow p & & \\ & p(z) & \\ h \nearrow & & \searrow p(\tilde{g}) \\ p(y) & \xrightarrow{p(\tilde{f})} & p(x) \end{array}$$

The reason for such name is because if we take the fiber at say $U \in C$ of p . Morphisms in P_U all maps to id_U , so P_U satisfies the property that outer horns Λ_0^2 and Λ_2^2 admits filler. This describes a groupoid (all isomorphisms).

The second property of the definition implies that the lifting of V given a lifting of U , x , along $f : V \rightarrow U$ exists, then it is unique up to isomorphism. Since suppose we have two lifting $y \rightarrow x$ and $y' \rightarrow x$, the outer horn they map to is filled with id_V , therefore, there exists an unique morphism $y \rightarrow y'$ and vice versa. Therefore, we may denote $y = f^*x$ and it is unique up to isomorphisms.

It seems like that given a category fibered in groupoid $p : P \rightarrow C$ determines a functor $\tilde{P} : C^{op} \rightarrow \mathbf{Grpd}$ such that $U \mapsto P_U$ and $(f : V \rightarrow U) \mapsto f^* : P_U \rightarrow P_V$ (when this makes sense).

This is not necessarily true. (Well first of all f^* might not be a well defined functor, but let's assume it is) The composition $(f^* \circ g^*) \circ h^*$ is only necessarily isomorphic to $f^* \circ (g^* \circ h^*)$. This tells us that $\tilde{P}$ is not a functor but a 2-functor. The study of category fibered in groupoids then is more difficult than the study of contravariant functors in groupoids.

Nevertheless, it is convenient to think of category fibered in groupoids as contravariant functors for us to see it as a generalization of presheaves of sets to presheaves of groupoids.

Definition 3.2. *Given a presheaf of groupoid $F : C^{op} \rightarrow \mathbf{Grpd}$, we call the Grothendieck construction of the presheaf to be the category fibered in groupoids associated to F .*

Example 3.1. *The forgetful functor $f : C/U = P \rightarrow C$ is fibered in groupoids, actually it is fibered in sets. The fiber at $V \in C$, $P_V = \text{Hom}_C(V, U)$. This tells us that representable presheaves are just overcategories. We call a category fibered in groupoids equivalent to C/U for some $U \in C$ representable.*

Definition 3.3. *Given two category fibered in groupoid $\mathcal{X}$ and $\mathcal{Y}$ over C , a morphism $f : \mathcal{X} \rightarrow \mathcal{Y}$ is representable if the for every $U \in C$ and $U \rightarrow \mathcal{Y}$, $U \times_{\mathcal{Y}} \mathcal{X}$ is again a representable.*

Before we introduce what a stack is, we need some technicalities.

Definition 3.4. *[9, 1.8.2] Given $p : P \rightarrow C$ fibered in groupoids, for any $U \in C$, and any $x, y \in P_U$, define functor*

$$\text{Mor}(x, y) : (C/U)^{op} \rightarrow \mathbf{Sets}$$

by

$$\text{Mor}(x, y)(f : V \rightarrow U) = \text{Hom}_{P_V}(f^*x, f^*y)$$

And for $h : (f : V \rightarrow U) \Rightarrow (g : W \rightarrow U)$ over U ,

$$\text{Mor}(x, y)(h) : \text{Hom}_{P_W}(g^*x, g^*y) \rightarrow \text{Hom}_{P_V}(f^*x, f^*y)$$

such that

$$\begin{aligned} \phi|_V &:= [\text{Mor}(x, y)(h)](\phi) \\ &= \alpha^{-1} \circ h^* \phi \circ \alpha \end{aligned}$$

Here α is the isomorphism between f^* and h^*g^* . So we are pulling a morphism $g^*x \rightarrow g^*y$ over P_W to a morphism $h^*g^*x \cong f^*x \rightarrow h^*g^*y \cong f^*y$ over P_V .

So information of the morphisms between x, y translate to a presheaf over U . For pullback, we denote $\pi_0 : X \times_Z Y \rightarrow X$ and $\pi_1 : X \times_Z Y \rightarrow Y$.

Definition 3.5. [9, Definition 1.8.3.5] Let $p : P \rightarrow C$ be fibered in groupoid, for $U \in C$ let $\mathcal{U} := \{f_i : U_i \rightarrow U\}$ be a set of morphisms over U . A descent datum (X_i, φ_{ij}) is a collection of object $X_i \in P_{U_i}$ and morphism $\varphi_{ij} : \pi_1^* X_i \rightarrow \pi_2^* X_j$ in $P_{U_i \times_U U_j}$ such that

$$\begin{array}{ccc} \pi_0^* X_i & \xrightarrow{\pi_{02}^* \varphi_{ik}} & \pi_2^* X_k \\ & \searrow \pi_{01}^* \varphi_{ij} \quad \nearrow \pi_{12}^* \varphi_{jk} & \\ & \pi_1^* X_j & \end{array}$$

commute in $P_{U_i \times_U U_j \times_U U_k}$

The meaning of this definition is a descent datum contains the information that we can glue together over $\mathcal{U}$. φ_{ij} identifies the information over the intersection $U_i \times_U U_j$ together (we cant ask for equality as it is unnatural for category theory), such that the overall identification on the intersection $U_i \times_U U_j \times_U U_k$.

A morphism of descent datum $\psi : (X_i, \varphi_{ij}) \rightarrow (X'_i, \varphi'_{ij})$ over $\mathcal{U}$ contains morphisms $\psi_i : X_i \rightarrow X'_i$ such that their pullback onto $U_i \times_U U_j$ is natural with φ_{ij} and φ'_{ij} .

This determines a category of descent datum of $\mathcal{U}$, $DD(\mathcal{U})$.

Definition 3.6. Given $\mathcal{U}$, and $X \in P_U$, the canonical descent datum of X respect to $\mathcal{U}$ is the $(f_i^* X, \text{can}_{ij})$, here can_{ij} is the canonical isomorphism between $\pi_0^* f_i^* X$ with $\pi_1^* f_j^* X$, because $f_i \circ \pi_0 = f_j \circ \pi_1$.

A descent datum (X_i, φ_{ij}) is effective if there is $X \in P_U$ such that (X_i, φ_{ij}) is isomorphic to the canonical descent datum of X .

Definition 3.7. [9, Definition 1.8.4.1] Let C be a site, $p : P \rightarrow C$ fibered in groupoid is a stack over C if

- For any $U \in C$, and $x, y \in P_U$, the presheaf $\text{Mor}(x, y) : (C/U)^{op} \rightarrow \text{Sets}$ is a sheaf over C/U .
- For every covering $\mathcal{U}$ of C , every descent datum of P respect to $\mathcal{U}$ is effective.

Both requirement tells us that for a stack, we can glue big groupoid over big object from a collection of small groupoids over a covering of small objects.

Definition 3.8. A groupoid object (X_0, X_1) in C is a functor $X_\bullet : \Delta_{\leq 2}^{op} \rightarrow C$, such that $(\text{Hom}_C(-, X_0), \text{Hom}_C(-, X_1))$ determines a presheaf of groupoid.

From now on, when we say stack over S , we mean a stack over $\mathcal{A}ff_S$ for some S , with the $fpqc$ -topology. If unspecified, assume $S = \text{spec}(\mathbb{Z})$

Definition 3.9. Given a representable morphism of stack over S , $f : \mathcal{M} \rightarrow \mathcal{M}'$, we say that f has property P , for some property of morphism between schemes, when for any affine $g : \text{spec}(R) \rightarrow \mathcal{M}'$, $g^* \mathcal{M} \cong \text{spec}(A) \rightarrow \text{spec}(R)$ has the property P (for instance, surjective, flat, faithfully flat, quasicompact, open immersion, ...).

Remark 3.2. Given a groupoid scheme (X_0, X_1) , we denote $\mathcal{M}_{(X_0, X_1)}$ stackification of the category fibered in groupoids associated with the presheaf $\text{Aff}_{/S}^{\text{op}} \rightarrow \text{Grpd}$ determined by (X_0, X_1) . There is a morphism $X_0 \rightarrow \mathcal{M}_{(X_0, X_1)}$. The pullback $X_0 \times_{\mathcal{M}} X_0 \cong X_1$, since X_1 represents the morphisms and the pullback describes the morphisms. If we keep going, we would have an augmented simplicial object

$$\dots \longrightarrow X_1 \times_{X_0} X_1 \rightrightarrows X_1 \rightrightarrows X_0 \longrightarrow \mathcal{M}_{(X_0, X_1)}$$

3.2 Quasicoherent Modules over Stack

Definition 3.10. Fix a stack over S , $\mathcal{M}$, we define $\text{Aff}_{/\mathcal{M}}$ with the fpqc-topology on $U \rightarrow \mathcal{M}$ for scheme U . The structure sheaf $\mathcal{O}_{\mathcal{M}} : \text{Aff}_{/\mathcal{M}}^{\text{op}} \rightarrow \text{Ring}$ is defined on the affines to be

$$\mathcal{O}_{\mathcal{M}}(\text{spec}(A) \rightarrow \mathcal{M}) = A$$

For morphism $f : \mathcal{M} \rightarrow \mathcal{M}'$, we define $f^{-1} \mathcal{O}_{\mathcal{M}'}(U \rightarrow \mathcal{M}) = \mathcal{O}_{\mathcal{M}}(U \rightarrow \mathcal{M} \rightarrow \mathcal{M}')$. There is $f^{-1} \mathcal{O}_{\mathcal{M}'} \rightarrow \mathcal{O}_{\mathcal{M}}$.

Definition 3.11. A quasicoherent sheaf over $\mathcal{M}$ is a $\mathcal{O}_{\mathcal{M}}$ -module $\mathcal{F}$ such that for $\text{spec}(B) \rightarrow \text{spec}(A)$ over $\mathcal{M}$

$$B \otimes_A \mathcal{F}(\text{spec}(A)) \rightarrow \mathcal{F}(\text{spec}(B))$$

is an isomorphism.

We denote the category of quasicoherent sheaf over $\mathcal{M}$ to be $\text{Qcoh}_{\mathcal{M}}$. Suppose we have morphism $f : \mathcal{M} \rightarrow \mathcal{M}'$, there exists adjoint pair

$$f^* : \text{Qcoh}_{\mathcal{M}'} \rightarrow \text{Qcoh}_{\mathcal{M}} : f_*$$

For $\mathcal{F} \in \text{Qcoh}_{\mathcal{M}}$, $f_* \mathcal{F}(V \rightarrow \mathcal{M}') = \mathcal{F}(f^* V \rightarrow \mathcal{M})$ and for $\mathcal{G} \in \text{Qcoh}_{\mathcal{M}'}$, $f^* \mathcal{G} = \mathcal{O}_{\mathcal{M}} \otimes_{f^{-1} \mathcal{O}_{\mathcal{M}'}} f^{-1} \mathcal{G}$, where $f^{-1} \mathcal{G}(U \rightarrow \mathcal{M}) = \mathcal{G}(U \rightarrow \mathcal{M} \rightarrow \mathcal{M}')$

If $\mathcal{M}$ is defined by groupoid (X_0, X_1) , how do determine the quasicoherent sheaf over $\mathcal{M}$ using (X_0, X_1) . Now if more X_0 and X_1 are affine, so they are equivalent to a Hopf-algebroid (A, Γ) . So for a qcoh module $\mathcal{F}$ over $\mathcal{M}$, do the simplicial stuff, if we let $\mathcal{F}(\text{spec}(A) \rightarrow \mathcal{M}) = M$

$$\dots \longleftarrow \Gamma \otimes_A \Gamma \otimes_A M \rightrightarrows \Gamma \otimes_A M \rightrightarrows M \longleftarrow \mathcal{F}(\mathcal{M})$$

Where $\mathcal{F}(\text{spec}(\Gamma) \rightarrow \mathcal{M}) = \Gamma \otimes_A M$ is determined by $d_0 : \text{spec}(\Gamma) \rightarrow \text{spec}(A)$. This identification let the morphism induced by d_1 be a coaction $M \rightarrow \Gamma \otimes_A M$.

So we have that $\mathcal{F} \in \text{Qcoh}_{\mathcal{M}}$ determines a (A, Γ) -comodules

Theorem 3.3. [2, Proposition 11.6] Let $\mathcal{M}$ be the stack associated with Hopf algebroid (A, Γ) ,

$$\text{Qcoh}_{\mathcal{M}} \simeq \text{coMod}_{(A, \Gamma)}$$

The cohomology is then super simple.

Proposition 3.4. *If $\text{spec}(A) \rightarrow \mathcal{M}$ defined by the domain is surjective and flat, for any $\mathcal{F} \in \mathcal{Q}\text{coh}_{\mathcal{M}}$, we have*

$$H^*(\mathcal{M}, \mathcal{F}) := \text{Ext}_{\mathcal{M}}^*(\mathcal{O}_{\mathcal{M}}, \mathcal{F}) \cong \text{Ext}_{(A, \Gamma)}^*(A, M)$$

4 Stack over Moduli Stack of Formal Groups

4.1 Groupoid Representation

Definition 4.1. *We define the moduli stack of formal groups $\mathcal{M}_{\text{fg}} \rightarrow \mathcal{A}\text{ff}$ to be the category fibered in groupoid with the following information:*

- *Objects are $((G, e), S)$ where $S \in \mathcal{A}\text{ff}$ and (G, e) is a formal group over S*
- *Morphisms $(\varphi, f) : ((H, e_H), T) \rightarrow ((G, e_G), S)$ contains a morphism of scheme $f : T \rightarrow S$ and an isomorphism $\varphi : (H, e_H) \rightarrow (f^*G, f^*e_G)$ of formal group over T .*

Recall that the ring L classifies formal group laws, there is morphism

$$\text{spec}(L) \rightarrow \mathcal{M}_{\text{fg}}$$

That takes the formal group law F over $\text{spec}(R)$ to the formal group determined F on $\text{spf}(R[[t]])$.

Theorem 4.1. *[1, Proposition 2.6] $\mathcal{M}_{\text{fg}}$ is a stack respects to $fpqc$ -topology.*

Given two formal group law F and G over R , the isomorphisms between F, G are invertible power series $\phi(t) = ut + b_1 t^2 + \dots$ with $u \in R^\times$ such that

$$\phi^{-1}F(\phi(x), \phi(y)) = G$$

The functor $\text{Isofgl} : \mathcal{A}\text{ff}^{op} \rightarrow \text{Sets}$ that takes $\text{spec}(R)$ to the sets (F, ϕ) where F is a formal group law over R and ϕ an invertible power series, or equivalently an isomorphism $\phi : F \rightarrow \phi(F)$.

The ring $W^+ := L[u^\pm, b_1, b_2, \dots]$ represents the presheaf Iso , so a $\varphi : W^+ \rightarrow R$ determines a formal group law from L and an invertible power series over R by $u^\pm, b_1, \dots$. This is actually a $fpqc$ -sheaf.

The pair $(\text{spec}(L), \text{spec}(W^+))$ determines a groupoid, where the $\text{dom} : \text{spec}(W^+) \rightarrow \text{spec}(L)$ is the domain of the formal group law F and $\text{cod} : \text{spec}(W^+) \rightarrow \text{spec}(L)$ is the codomain $\phi(F)$. Let $\mathcal{M}_{(L, W^+)}$ be the stack associated to $(\text{spec}(L), \text{spec}(W^+))$.

It is not hard to believe that there exists an equivalence between $\mathcal{M}_{(L, W^+)} \rightarrow \mathcal{M}_{\text{fg}}$.

4.2 Localization at Evenly Periodic Cohomology

A cohomology E is called evenly periodic if there exists a unit $\beta \in E^2$ and $E^1 \cong 0$. So all odd degree are zero and even degrees looks the same, $E^* \cong \pi_0 E[\beta^\pm]$. Obviously, from ATSS, E is complex oriented.

We may turn MP into an evenly periodic cohomology by simply tensor it with $\mathbb{Z}[\beta^\pm]$. We refer to the resulting cohomology MP .

Given the formal group law of E , we have that $F(\beta^{-1}x, \beta^{-1}y)$ is a formal group law over $\pi_0 E$. Hence a $f : \text{spec}(\pi_0 E) \rightarrow \mathcal{M}_{\text{fg}}$. Recall for every ring spectrum, there is a cosimplicial spectrum $\mathcal{F}_\bullet^E X := E^{\wedge \bullet + 1} \wedge X$. We have that $\pi_0 \mathcal{F}_\bullet^E S$ is a cosimplicial ring and a simplicial affine map $\text{spec } \pi_0 \mathcal{F}_\bullet^E S \rightarrow \mathcal{M}_{\text{fg}}$.

If we extend further, making some assumption on E

Definition 4.2. [6] We say E satisfies commutative hypothesis CH if $\mathcal{F}_\bullet^E S$ are all even periodic, and say E satisfy flat hypothesis if $FH E_0 \rightarrow E_0 E$. In this case, $\pi_0 \mathcal{F}_n^E X \cong E_0 E^{\otimes_{E_0} n} \otimes_{E_0} E_0 X$.

Therefore, we have for E satisfy CH and FH, $(E_0, E_0 E)$ is a Hopf algebroid and let $\mathcal{M}_E$ be the stack associated to E , and let $\mathcal{M}_E(X)$ be the qcoh module associated with $\pi_0 \mathcal{F}_\bullet^E X$.

We have that $\text{Ext}_{(E_0, E_0 E)}^s(E_0, E_0 X) \cong \text{Ext}_{(E_*, E_* E)}^s(E_*, E_* X)$ by the periodicity, and there is spectral sequence

$$E_2^{s, 2t} = \text{Ext}_{(E_*, E_* E)}^{s, 2t}(E_*, E_* X) \cong H^s(\mathcal{M}_E, \mathcal{M}_E(X) \otimes \omega_E^t) \implies \pi_{2t-s} L_E X$$

From ANSS [6].

Example 4.2. Let $E = MP$, then since $MP_0 \cong L$ and $MP_0 MP \cong W^+$, $\mathcal{M}_{MP} \cong \mathcal{M}_{\text{fg}}$ and

$$H^s(\mathcal{M}_{\text{fg}}, \omega_{\text{fg}}^t) \implies \pi_{2t-s} L_{MP} S \cong \pi_{2t-s} S$$

Since MP -acyclics are S -acyclic.

So we have went from homotopy to algebraic geometry.

5 Formal Groups over $\mathbb{Z}_{(p)}$ -Schemes

5.1 Height Filtration of $\mathcal{M}_{fg} \otimes \mathbb{Z}_{(p)}$

‘ Let p be some prime, we let $\mathcal{M}_{\text{fg}} := \mathcal{M}_{\text{fg}} \otimes \mathbb{Z}_{(p)}$ be the stack of formal group over $\text{Aff}_{/\text{spec } \mathbb{Z}_{(p)}}$.

Definition 5.1. Let F be a formal group law over a p -local ring R , let v_n be the coefficient of the term t^{p^n} in the power series $[p]_F(t)$ (obviously $v_0 = p$), we say F has height $\geq n$ if $v_i = 0$ for $i < n$, (so if F has height ≥ 1 , R has characteristic p), and F has height exactly n if it has height $\geq n$ and v_n is invertible. If $[p]_F t = 0$ we say $\text{hgt } F = \infty$.

Given two formal group law F, G over R , and an isomorphism $\phi : F \rightarrow G$. We have that

$$[p]_F(t) = (\phi^{-1} \circ [p]_G \circ \phi)(t)$$

And $\text{hgt} F \geq n$ or $\text{hgt} F = n$ iff $\text{hgt} G \geq n$ or $\text{hgt} G = n$. Therefore, we have that height is actually an invariant of the formal group.

We define $\mathcal{M}_{\text{fg}}^{\geq n}$ to be the closed substack of formal group with $\text{hgt} \geq n$. There is then inclusion

$$\dots \subseteq \mathcal{M}_{\text{fg}}^{\geq n} \subseteq \dots \subseteq \mathcal{M}_{\text{fg}}^{\geq 1} \subseteq \mathcal{M}_{\text{fg}}^{\geq 0} = \mathcal{M}_{\text{fg}}$$

Example 5.1. $\mathcal{M}_{\text{fg}}^{\geq 1}$ classifies the formal group law such that $p = 0$, which is equivalent to $\mathcal{M}_{\text{fg}} \otimes \mathbb{F}_p$.

We denote $\mathcal{H}(n) := \mathcal{M}_{\text{fg}}^{\geq n} - \mathcal{M}_{\text{fg}}^{\geq n+1}$ the open substack of $\mathcal{M}_{\text{fg}}^{\geq n}$ of the complement of $\mathcal{M}_{\text{fg}}^{\geq n+1}$ (the maximal open immersion $\mathcal{X} \rightarrow \mathcal{M}_{\text{fg}}^{\geq n}$ disjoint from $\mathcal{M}_{\text{fg}}^{\geq n+1}$).

Example 5.2. $\mathcal{M}_{\text{fg}}^{\geq 1}$ classifies the formal group law such that $p = 0$, which is equivalent to $\mathcal{M}_{\text{fg}} \otimes \mathbb{F}_p$. $\mathcal{M}_{\text{fg}}^0 \cong \mathcal{M}_{\text{fg}} \otimes \mathbb{Q} \cong B\mathbb{G}_m^{\mathbb{Q}}$.

Let $I_n = (p, v_1, \dots, v_{n-1})$, there exists faithfully flat

$$\begin{aligned} \text{spec } L_{(p)} &\rightarrow \mathcal{M}_{\text{fg}} \\ \text{spec}(L_{(p)}/I_n) &\rightarrow \mathcal{M}_{\text{fg}}^{\geq n} \\ \text{spec}(v_n^{-1}L_{(p)}/I_n) &\rightarrow \mathcal{H}(n) \end{aligned}$$

As it turns out, p -locally, for every formal group, we can choose a coordinate and a formal group law F over this coordinate such that

$$[p]_F t = pt + v_1 t^p + v_2 t^{p^2} + \dots$$

It is not hard to believe that there is a better groupoid representation of the stack we talked about. Let $B = \mathbb{Z}_{(p)}[v_1, v_2, \dots]$ and let $I_n = (p, v_1, \dots, v_{n-1})$

$$\begin{aligned} \text{spec}(B) &\rightarrow \mathcal{M}_{\text{fg}} \\ \text{spec}(B/I_n) &\rightarrow \mathcal{M}_{\text{fg}}^{\geq n} \\ \text{spec}(v_n^{-1}B/I_n) &\rightarrow \mathcal{H}(n) \end{aligned}$$

For those of us who knows the p -localization of MU , we recall that

$$\begin{aligned} L_{(p)}MU &\simeq \bigvee_{i \in I} \Sigma^i BP \\ BP_* &\cong \mathbb{Z}_{(p)}[v_1, v_2, \dots], |v_i| = 2(p^i - 1) \end{aligned}$$

Here BP is the Brown-Peterson spectrum, whose homology is $BP_*X \cong MU_*X \otimes_{MU_*} BP_*$ (this makes sense since BP_* is Landweber exact). Specifically, we have

$$BP_*BP[u^{\pm}] \cong BP_* \otimes_{MU_*} MU_*MU \otimes_{MU_*} BP_*[u^{\pm}] \cong \text{spec}(BP_*) \times_{\mathcal{M}_{\text{fg}}} \text{spec}(BP_*)$$

, so we have $\mathcal{M}_{\text{fg}}^{\geq n}$ is associated with $(BP_*, BP_*BP[u^\pm])$ and

$$H^s(\mathcal{M}_{\text{fg}} \otimes_{\mathbb{Z}_{(p)}} \omega^t \otimes \mathbb{Z}_{(p)}) \implies \pi_{2t-s} L_{(p)} S$$

Let $\mathcal{X}(n)$ to be the open complement of $\mathcal{M}_{\text{fg}}^{\geq n+1}$, since $\mathcal{X}(n) \times_{\mathcal{M}_{\text{fg}}} \text{spec}(BP_*) \cong \text{spec}(\mathbb{Z}_{(p)}[v_1, \dots, v_{n-1}, v_n^\pm])$, there is a flat covering

$$\text{spec}(E(n)_*) := \text{spec}(\mathbb{Z}_{(p)}[v_1, \dots, v_{n-1}, v_n^\pm]) \rightarrow \mathcal{X}(n)$$

Since $\mathcal{X}(n)$ is open, it is every much flat, and we have homology $E(n)_* X = BP_* X \otimes_{BP_*} E(n)_*$, with $E(n)_* E(n) \cong E(n)_* \otimes_{BP_*} \otimes_{BP_*} BP \otimes_{BP_*} E(n)_*$. This is the Johnson-Wilson spectrum.

We leave out the discussion on the homotopic properties of $E(n)$. However, there is one coming from the geometric property of $\mathcal{X}(n)$.

For any $X \in \mathcal{S}$, let $\mathcal{F}_X$ be the qcoh module over $\mathcal{M}_{\text{fg}}$ associated with $BPP_0 X$ (periodic version), we have that $E(n)P_0 X \cong i_n^* \mathcal{F}_X$ the pullback along inclusion. We see that if for any $\mathcal{F} \in \mathcal{Q}\text{coh}_{\mathcal{M}}$, $i_n^* \mathcal{F} = 0$ then $i_{n-1}^* \mathcal{F} = 0$. For $\mathcal{F}_X$, this implies that $E(n)$ -acyclic are $E(n-1)$ -acyclic, equivalently $E(n-1)$ -local are $E(n)$ -local, and we have a tower of localization functor

$$L_{(p)} \rightarrow \dots \rightarrow L_n := L_{E(n)} \rightarrow L_{n-1} \rightarrow \dots \rightarrow L_1 \rightarrow L_0 \cong L_{\mathbb{Q}}$$

This is the Chromatic Tower.

5.2 Fracture and Chromatic Convergence Theorem

We now work a little bit in the derived setting (just a little). Given a stack $\mathcal{M}$, let $\mathcal{D}(\mathcal{M})$ be the derived category of $\mathcal{Q}\text{coh}_{\mathcal{M}}$.

We say that a qcoh module $\mathcal{F}$ over $\mathcal{M}$ with a Hopf algebroid (A, Γ) is coherent if $\mathcal{F}(A)$ is a finitely presented A -module.

Theorem 5.3. [1, Theorem 8.22] (Algebraic Chromatic Convergence) For $\mathcal{F}$ coherent over $\mathcal{M}_{\text{fg}}$,

$$\mathcal{F} \rightarrow \mathbb{R} \lim_n [\mathbb{R}(i_n)_*] i_n^* \mathcal{F}$$

is an equivalence.

The open immersion of $\mathcal{H}(n)$ in $\mathcal{X}(n)$ is defined by the loci $(p, v_1, \dots, v_{n-1})$ in $\mathcal{X}(n)$.

Definition 5.2. [1] Let $\mathcal{X} \subseteq \mathcal{M}$ be a closed substack with defining ideal $\mathcal{I} \subseteq \mathcal{O}_{\mathcal{M}}$, we then define

$$L_{\mathcal{X}}(\mathcal{F}) := \mathbb{R} \lim_n (\mathcal{O}_{\mathcal{M}} / \mathcal{I}^n \otimes^{\mathbb{L}} \mathcal{F})$$

Suppose we let $i_n : \mathcal{X}_n \rightarrow \mathcal{M}$ the closed substack determined by $\mathcal{I}^n$, then $L_{\mathcal{X}}(\mathcal{F}) \simeq \mathbb{R} \lim_n (i_n)_* \mathbb{L}(i_n)^* \mathcal{F}$.

We here assume that $\mathcal{Y} \rightarrow \mathcal{M}_{\text{fg}}$ to be quasicompact and separated, then the open substack filtration $\mathcal{X}(n) \subseteq \mathcal{M}_{\text{fg}}$ induce a open filtration $i_n : \mathcal{Y}(n) \rightarrow \mathcal{Y}(n+1) \subseteq \mathcal{Y}$. We denote $\mathcal{Z}(n) = \mathcal{H}(n) \times_{\mathcal{X}(n)} \mathcal{Y}(n)$ the closed substack.

Theorem 5.4. [1, Theorem 8.17] (Fracture Theorem) Let $\mathcal{F} \in \mathcal{Q}\mathrm{cohy}_{(n)}$ be quasicoherent

$$\begin{array}{ccc} \mathcal{F} & \xrightarrow{\quad\quad\quad} & L_{\mathcal{Z}(n)}(\mathcal{F}) \\ \downarrow & & \downarrow \\ [\mathbb{R}(i_{n-1})_*](i_{n-1})^* \mathcal{F} & \longrightarrow & [\mathbb{R}(i_{n-1})_*](i_{n-1})^* L_{\mathcal{Z}(n)}(\mathcal{F}) \end{array}$$

is a homotopy pullback.

See [1, Section 8] for a detailed discussion, here we care about what it tells us about homotopy.

Remark 5.5. This is just my discussion on why we can expect chromatic convergence in homotopy (of course, none should be taken rigorously). For algebraic chromatic convergence, we first have that

$$0 \rightarrow \lim_n^1 H^{s+1}(\mathcal{X}(n), i_n^* \mathcal{F}) \rightarrow H^s(\mathcal{M}_{\mathrm{fg}}, \mathcal{F}) \rightarrow \lim_n H^s(\mathcal{X}(n), i_n^* \mathcal{F}) \rightarrow 0$$

Let $\mathcal{F} = \mathcal{F}_{\Sigma^{-t}X} \otimes \omega^t$ for finite p -local spectrum, which is then coherent, we would then expect an exact sequence from the localization spectral sequence

$$0 \rightarrow \lim_n^1 \pi_{2t-s-1}(L_n X) \rightarrow \pi_{2t-s}(L_{(p)} X) \rightarrow \lim_n \pi_{2t-s}(L_n X) \rightarrow 0$$

Which we then would expect $L_{(p)} X \simeq \mathrm{holim} L_n X$.

Remark 5.6. The discussion of $L_{\mathcal{Z}(n)}(\mathcal{F})$ should probably be left until we talk about $K(n)$, the Morava K -theory, which $K(n)_* \cong \mathbb{F}_p[v_n^\pm]$. The localization $L_{K(n)}$ of some $E(n)$ -local spectrum is equivalent to the completion of qcoh module over $\mathcal{X}(n)$, respect to the ideal of definition $\mathcal{H}(n) \rightarrow \mathcal{X}(n)$, which is $(p, v_1, \dots, v_{n-1})$.

6 Height Exactly at n

6.1 Groupoid Representation $\mathcal{H}(n)$

For $n \geq 1$, we have that $\mathcal{H}(n)$ is essentially over characteristic p .

Theorem 6.1. [1] (Lazard Uniqueness Theorem) Given a field of characteristic p , $\mathbb{F}$, for every two formal group law F, G , of height exactly n , there is a separable extension $i : \mathbb{F} \rightarrow L$, such that there is an isomorphism between $i^* F$ and $i^* G$.

Therefore, we have an unique formal group law Γ of height exactly n over $\overline{\mathbb{F}}_p$ and a faithfully flat map

$$\mathrm{spec}(\overline{\mathbb{F}}_p) \rightarrow \mathcal{H}(n)$$

And we have the group scheme $\mathcal{A}\mathrm{ut}(\overline{\mathbb{F}}_p, \Gamma) := \mathrm{spec}(\overline{\mathbb{F}}_p) \times_{\mathcal{H}(n)} \mathrm{spec}(\overline{\mathbb{F}}_p)$ that takes a $\mathbb{F}_p$ -scheme $f : S \rightarrow \mathrm{spec}(\mathbb{F}_p)$ to the automorphism of $i^* \Gamma$ over S .

The $\mathbb{F}_p$ points $\mathbb{G}_n$ sits inside a split exact sequence

$$0 \rightarrow SS_n \rightarrow \mathbb{G}_n \rightarrow \mathrm{Gal}(\overline{\mathbb{F}}_p/\mathbb{F}_p) \rightarrow 0$$

Here SS_n is the $\overline{\mathbb{F}}_p$ -points of $\mathcal{A}\mathrm{ut}(\overline{\mathbb{F}}_p, \Gamma)$. So $\mathcal{H}(n) \cong [\mathrm{spec}(\overline{\mathbb{F}}_p)/\mathbb{G}_n]$

6.2 Lubin-Tate Deformation as a Universal Cover

Recall that $\mathcal{H}(n) \subseteq \mathcal{X}(n)$ is a closed substack, we denote $\widehat{\mathcal{H}(n)}$ the infinitesimal formal neighbourhood of $\mathcal{H}(n)$ in $\mathcal{X}(n)$. If we wish to study $\mathcal{X}(n)$ from $\mathcal{X}(n-1)$ and $\mathcal{H}(n)$, we would need to know how they intersect, hence studying $\widehat{\mathcal{H}(n)}$.

In topology, when we study a space, we would often like to look at its universal cover, because they only differ by a group action and the latter is much well behaved.

Definition 6.1. Let κ be some perfect field of characteristic p , we denote Art_κ the category of Artin local rings $(A, \mathfrak{m})$ where $\kappa \cong A/\mathfrak{m}$ and morphisms are ring homomorphisms.

Suppose we have a formal group of height n Γ over κ , we define category fibered in groupoid $\text{Def}(\kappa, \Gamma)$ over $\text{Art}_\kappa^{\text{op}}$, with objects

$$\begin{array}{ccc} \Gamma & \longrightarrow & F \\ \downarrow & & \downarrow \\ \text{spec}(\kappa) & \xrightarrow{i} & \text{spec}(A) \end{array}$$

such that F is a formal group over $\text{spec}(A)$, i determines an isomorphism of $\text{spec}(\kappa) \cong \text{spec}(A/\mathfrak{m})$, and an isomorphism $i^*F \cong \Gamma$.

A morphism from $(\text{spec}(A), F)$ to $(\text{spec}(B), G)$ is then a diagram

$$\begin{array}{ccccc} & & & & G \\ & & & \nearrow & \downarrow \\ \Gamma & \longrightarrow & F & \longrightarrow & \text{spec}(B) \\ \downarrow & & \downarrow & \nearrow f & \\ \text{spec}(\kappa) & \xrightarrow{i} & \text{spec}(A) & & \end{array}$$

and an isomorphism $f^*G \cong F$. A geometric way to think of $\text{Def}(\kappa, \Gamma)$ is to see it as all the thickening of κ and Γ .

Example 6.2. Let $\kappa = \mathbb{F}_p$ and $\Gamma = x + y + xy$, we know that $\text{hgt } \Gamma = 1$. Let $A = \mathbb{Z}_{(p)}$ the maximal ideal is (p) , and $A/(p) \cong \kappa$. We may also let $F = x + y + xy$. Note, (A, F) is a deformation of (κ, Γ) , but $\text{hgt } F = 0$. So we have that $\text{Def}(\kappa, \Gamma)$ contains more information than simply height n .

Let R be a complete local ring $R/\mathfrak{m} \cong \kappa$, we denote $\text{spf}(R)$ the category fibered in groupoid over Art_κ that takes $(A, \mathfrak{m})$ to ring map $f : R \rightarrow A$ such that $R/\mathfrak{m} \cong A/\mathfrak{m}$.

The category $\text{Def}(\kappa, \Gamma)$ is actually discrete.

Theorem 6.3. [1, Theorem 7.4] (Lubin-Tate Theorem) There exists a complete local ring $R(\kappa, \Gamma)$, a formal group H over $R(\kappa, \Gamma)$ and a deformation of $(\text{spf}(R(\kappa, \Gamma)), H)$ such that $\text{spf}(R(\kappa, \Gamma)) \rightarrow \text{Def}(\kappa, \Gamma)$ is an equivalence of category fibered in groupoids over Art_κ .

Actually, $R(\kappa, \Gamma) \cong W(\kappa)[[v_1, \dots, v_{n-1}]]$, to determine all possible information of height $\leq n$. Here $W(\kappa)$ is the ring of Witt vectors of κ .

There is a fully faithful forgetful functor $\text{Art}_\kappa^{op} \rightarrow \mathcal{A}ff$ that takes $(A, \mathfrak{m})$ to $\text{spec}(A)$. We then extend $\text{Def}(\kappa, \Gamma)$ over the entire $\mathcal{A}ff$ through left Kan extension, that is for $B \in \mathcal{A}ff$

$$\text{Def}(\kappa, \Gamma)(B) := \text{colim}_{A \in \text{Art}_\kappa / B} \text{Def}(\kappa, \Gamma)(A)$$

So a deformation of $\text{spec}(B) \in \mathcal{A}ff$ contains a $f : \text{spec}(B) \rightarrow \text{spec}(A)$ to some Artin local ring A and a deformation over it.

There is a map $\text{Def}(\kappa, \Gamma) \rightarrow \widehat{\mathcal{H}(n)}$ that takes a deformation $(f : \text{spec}(B) \rightarrow \text{spec}(A), F)$ of Γ over $\text{spec}(B)$ and takes it to f^*F over $\text{spec}(B)$. The map is representable and faithfully flat.

The group $\mathbb{G}_n$ act upon $\text{Def}(\overline{\mathbb{F}}_p, \Gamma)$ from the following, let $(\sigma, g) \in \mathbb{G}_n$ where σ is the action of Gal and g is an isomorphism of Γ , so we have

$$\begin{array}{ccccc} \Gamma & \xrightarrow{g} & \Gamma & \longrightarrow & F \\ \downarrow & & \downarrow & & \downarrow \\ \text{spec}(\overline{\mathbb{F}}_p) & \xrightarrow{\sigma} & \text{spec}(\overline{\mathbb{F}}_p) & \xrightarrow{i} & \text{spec}(A) \end{array}$$

Remark 6.4. The group $\mathbb{G}_n$ is actually profinite, we describe it here.

A k -buds of formal group law over R is a power series $F(x, y) \in R[[x, y]]/(x, y)^{k+1}$ satisfying the properties of a formal group law (of course up to $\text{mod } (x, y)^{k+1}$) and an isomorphism is then a power series $\phi(t) \in R[[t]]/(t)^{k+1}$ such that $\phi(F(x, y)) \equiv F(\phi(x), \phi(y))$.

If we have a formal group law F over R , then $F \text{ mod } (x, y)^{k+1}$ is a k -bud, similarly, an isomorphism of formal group law is then an isomorphism of k -buds $\text{mod } t^{k+1}$.

We define subgroup $SS_n^i \subseteq SS_n$ to be the normal subgroup containing automorphism g that equals to the identity of Γ on p^i -bud. We then have

$$SS_n \cong \lim SS_n / SS_n^i$$

Since Gal is also profinite, we define the profinite structure $\mathbb{G}_n^i$ on $\mathbb{G}_n$ from both SS_n and Gal .

This extends to a profinite action of $\mathbb{G}_n$ on $\text{Def}(\overline{\mathbb{F}}_p, \Gamma)$. We define $\text{Def}(\overline{\mathbb{F}}_p, \Gamma)_i$ to be the groupoid of $(G, i, \text{spec}(A))$, where G is still a formal group over $\text{spec}(A)$, but we only require isomorphism of i^*G with Γ on the p^i -buds. Since isomorphisms on buds for local rings lift to an isomorphism

$$\text{Def}(\overline{\mathbb{F}}_p, \Gamma) \rightarrow \text{Def}(\overline{\mathbb{F}}_p, \Gamma)_i$$

is surjective. Moreover, both have action of $\mathbb{G}_n$ and the map is a $\mathbb{G}_n$ -map.

The action of $\mathbb{G}_n$ on $\text{Def}(\overline{\mathbb{F}}_p, \Gamma)_i$ factors through $\mathbb{G}_n / \mathbb{G}_n^i$ and

$$\text{Def}(\overline{\mathbb{F}}_p, \Gamma) \xrightarrow{\cong} \lim \text{Def}(\overline{\mathbb{F}}_p, \Gamma)_i$$

We have a profinite $\mathbb{G}_n$ -structure on $\text{Def}(\overline{\mathbb{F}}_p, \Gamma)$.

This action translate to an action on $R(\kappa, \Gamma)$.

As it turns out, we have

$$\text{Def}(\kappa, \Gamma) / \mathbb{G}_n \xrightarrow{\cong} \widehat{\mathcal{H}(n)}$$

Lastly, $R(\kappa, \Gamma)$ is Landweber exact, so there is a homology

$$E(\kappa, \Gamma)_* X = MP_* X \otimes_{MP_0} R(\kappa, \Gamma)$$

This is called Lubin-Tate E -theory, the localization at $E(\kappa, \Gamma)$ is the n^{th} chromatic layer L_n .

6.3 Devinatz-Hopkins Spectral Sequence

I think I should end this with something that I really like.

Let E_n be the Lubin-Tate E -theory at height n . We define

$$(E_n)_*^\vee E_n := \pi_* L_{K(n)}(E_n \wedge E_n)$$

A classic result of the above [3]

$$(E_n)_*^\vee E_n \cong C_{cts}(\mathbb{G}_n, (E_n)_*)$$

The continuous functions with respect to the profinite topology on $\mathbb{G}_n$ and $(E_n)_*$.

Of course, this identification goes further, as it turns out [2]

$$\begin{aligned} \pi_* L_{K(n)}(E_n^{\wedge^{i+1}} \wedge X) &\cong ((E_n)_*^\vee E_n)^{\hat{\otimes}^n} \hat{\otimes}_{E_*} E_* X \\ &\cong C_{cts}(\mathbb{G}_n^{\times i}, (E_n)_* X) \end{aligned}$$

It should not be surprising that we can put them in a Bousfield Kan spectral sequence, with E_2 page the continuous cohomology $H_{cts}^*(\mathbb{G}_n, (E_n)_* X)$ and converge to the homotopy of totalization of $L_{K(n)}(E_n^{\wedge^{\bullet+1}} \wedge X)$. By [2, Corollary A.8], if X is $K(n)$ -local, this is equivalent to X itself, and since $K(n)$ implies $E(n)$ -local which is equivalent to E_n -local, we have for any $X \in \mathcal{S}$, a spectral sequence

$$E_2^{s,t} = H_{cts}^s(\mathbb{G}_n, (E_n)_t X) \implies \pi_{t-s} L_{K(n)} X$$

This is called the Devinatz-Hopkins spectral sequence.

Remark 6.5. *The first appearance of this spectral sequence comes from the study of the E_2 page of ANSS of $K(n)$, which is equivalent to H_{cts}^* , see [7].*

This also comes from the interpretation of ANSS for $\widehat{\mathcal{H}(n)}$.

We expect a spectral sequence

$$E_2^{s,t} = H^s(\widehat{\mathcal{H}(n)}, \mathcal{F}_{\Sigma^{-t}X} \otimes \omega^t) \implies \pi_{2t-s} L_{\widehat{\mathcal{H}(n)}} X$$

We expect $L_{\widehat{\mathcal{H}(n)}} X = L_{K(n)}$. If we pullback $\mathcal{F}_{\Sigma^{-t}X} \otimes \omega^t$ onto $\text{Def}(\kappa, \Gamma)$, the qcch correspond to $(E_n)_t X$ (of course with the profinite structure considered). Lastly, we have $\widehat{\mathcal{H}(n)} \cong \text{Def}(\kappa, \Gamma)/\mathbb{G}_n$, so the cohomology will be the fix point cohomology (respect to the profinite topology). This recovers the Devinatz-Hopkins spectral sequence.

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